

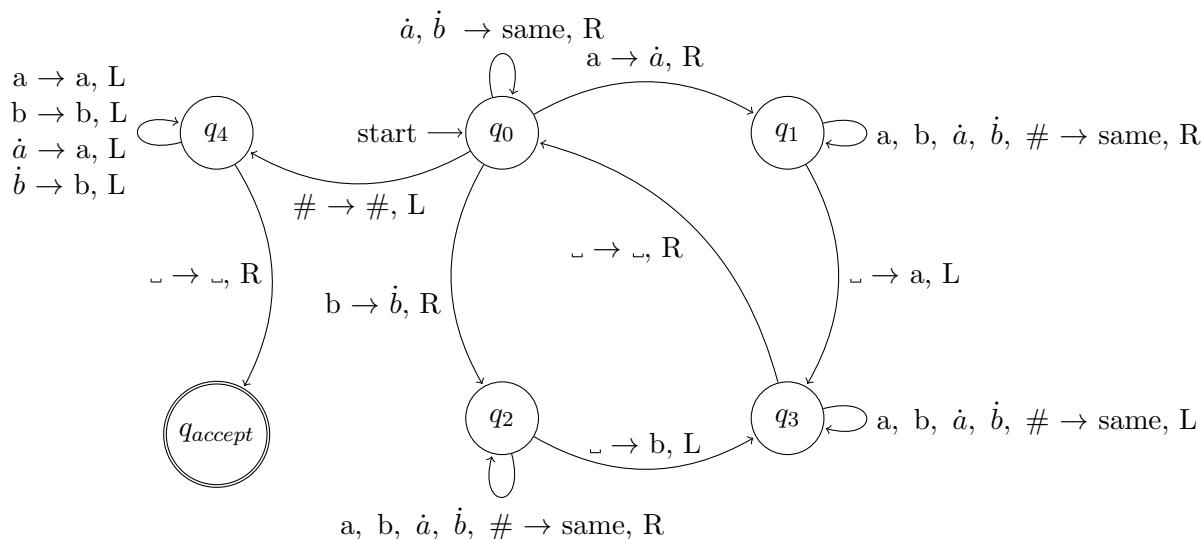
Homework 5: Turing Machines

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You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments. Three bonus points if you use L^AT_EX, and our template. You may collaborate with other students, but any written work should be your own.

1. Give the state diagram of a Turing machine which begins with $w\#$ on its tape for any $w \in \Sigma^*$ and halts with $w\#w$ on its tape. The input alphabet is $\Sigma = \{a, b\}$ and the tape alphabet may be $\Gamma = \{a, b, \dot{a}, \dot{b}, \#, \sqcup\}$. This Turing machine performs the copy ability. We gave a decision version of this in class. You are giving the computation version.



2. Give the sequence of configurations of your state diagram from question 2 beginning from $q_0aba\#$. Each on a new line please. (Hint, why not write a program for this)
 - (a) $q_0 a b a \# \sqcup \sqcup \dots$
 - (b) $\dot{a} q_1 b a \# \sqcup \sqcup \dots$
 - (c) $\dot{a} b q_1 a \# \sqcup \sqcup \dots$
 - (d) $\dot{a} b a q_1 \# \sqcup \sqcup \dots$
 - (e) $\dot{a} b a q_1 \# \sqcup \sqcup \dots$ [Head on blank]
 - (f) $\dot{a} b a q_3 \# a \sqcup \dots$ [Write copy of a]
 - (g) $\dot{a} b q_3 a \# a \sqcup \dots$ [Moving left]
 - (h) $\dot{a} q_3 b a \# a \sqcup \dots$

- (i) $\sqcup q_3 \dot{a} b a \# a \sqcup \dots$ [Continue left until blank]
- (j) $\sqcup q_0 \dot{a} b a \# a \sqcup \dots$ [Upon blank, move right into q_0]
- (k) $\sqcup \dot{a} q_0 b a \# a \sqcup \dots$
- (l) $\sqcup \dot{a} \dot{b} q_2 a \# a \sqcup \dots$ [Mark b]
- (m) $\sqcup \dot{a} \dot{b} a q_2 \# a \sqcup \dots$
- (n) $\sqcup \dot{a} \dot{b} a \# q_2 a \sqcup \dots$
- (o) $\sqcup \dot{a} \dot{b} a \# a q_2 \sqcup \dots$
- (p) $\sqcup \dot{a} \dot{b} a \# a q_3 b \sqcup \dots$ [Copy b]
- (q) $\sqcup \dot{a} \dot{b} q_3 a \# a b \sqcup \dots$ [Moving left]
- (r) $\sqcup q_3 \dot{a} \dot{b} a \# a b \dots$
- (s) $\sqcup q_0 \dot{a} \dot{b} a \# a b \dots$ [Back at left end]
- (t) $\sqcup \dot{a} \dot{b} q_0 a \# a b \dots$
- (u) $\sqcup \dot{a} \dot{b} q_1 \dot{a} \# a b \dots$ [Mark third letter a]
- (v) $\sqcup \dot{a} \dot{b} \dot{a} \# q_1 a b \dots$
- (w) $\sqcup \dot{a} \dot{b} \dot{a} \# a q_1 b \dots$
- (x) $\sqcup \dot{a} \dot{b} \dot{a} \# a b q_1 \sqcup \dots$
- (y) $\sqcup \dot{a} \dot{b} \dot{a} \# a b q_3 a \dots$ [Copy third letter a]
- (z) $\sqcup \dot{a} \dot{b} \dot{a} q_4 \# a b a \dots$ [Cleanup phase starts]
- () $q_{accept} a b a \# a b a \dots$

3. Give a high level, detailed, description of a Turing machine which computes the projection function $U(n, i, x_1, x_2, \dots, x_n) = x_i$. Do not give a state diagram. The Turing machine must begin with $1^n \# 1^i \# x_1 \# \dots \# x_n$ and halt with just x_i on its tape left-shifted fully. If this was psuedocode, the Turing machine would compute the following algorithm:

```
def U(n, i, A[]):
    return A[i]
```

We recognize the Turing Machine's input tape is formatted such that it is in the form of $1^n \# 1^i \# x_1 \# \dots \# x_n$, where $1^n, 1^i$ are unary representations of n, i . We wish for the machine to skip over the first 1^n cells and the first delimiter. We then utilize 1^i as a counter, where we process each 1 by skipping over one x-block. Therefore, after consuming all 1's in 1^i , we find the head will arrive at the beginning of the i -th block (or x_i). The machine will then copy x_i into blank cells located at the leftmost end of the tape. This will erase any remaining input cells and shift the copied x_i to the left. We observe the machine will halt only when x_i is on the tape. We have arrived at our solution.

4. Prove that the computable functions are closed under composition.

We wish to assume we have two functions f, g which are computable with corresponding Turing Machines M_1, M_2 that compute them. By definition, this means on any input x , M_1 will eventually halt with $f(x)$ on its tape and, similarly, on any input y , M_2 will eventually halt with $g(y)$ on its tape. To prove computable functions are closed under composition, we wish to show there exists a function h that is defined as $h(x) = g(f(x))$.

We wish to construct a new Turing Machine that, when given an input x , first simulates M_1 on x , thereby computing $f(x)$. Once the Turing Machine halts, we wish to use the output as input for a simulation of M_2 . We find that since M_1, M_2 are guaranteed to halt on every input that is found in their domain, M will eventually halt with $g(f(x))$ on its tape. Therefore, we effectively have chained the computations, thereby producing h (which is therefore computable). We observe, as a result, the set of computable functions is closed under composition.